Finding and Filling Gaps in the Literature

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***Abstract*—Language models are increasingly being used to generate long-form pedagogical content, including textbooks. We consider a prerequisite problem of identifying the topics that are in need of textbooks. We propose a geometric approach, using denoised document embeddings derived from the titles and abstracts of articles and books from the research literature. A *gap* is a region in this embedding space with many articles but no books, and hence a candidate textbook topic. For 73 of 76 topics within the AI literature, our identified gaps are larger than every gap filled by a published book on the topic. We demonstrate the feasibility of filling these gaps by having an LM generate a book proposal (title and abstract) for each gap. The median synthetic proposal is closer than any book to 96.4% of in-gap articles, and in a blind survey of 22 participants the proposals were statistically indistinguishable from existing books on the same topics across all three criteria (interesting, well-written, publishable).**

***Index Terms*—Data mining, Anomaly detection, Nearest neighbor methods, Generative AI.**

I. Introduction

Language models are being applied to increasingly ambitious projects, from writing research papers [1] to the on-the-fly generation of interactive worlds [2]. One promising area of application is in generating pedagogical content, where AI models are being applied to elaborating short overviews of a topic into educational deep-dives [3], virtual peers and teachers [4], and end-to-end course content creation (including a textbook and quizzes) extracted from uploaded materials [5].

A particular challenge is that of identifying which topics are in genuine need of books, and then synthesizing books to fill these gaps. There is increasing anecdotal evidence that published books are now being written by AI [6]. By one estimate, 15% of the self-published genre-fiction e-books released on Amazon from 2023 to early 2026 are mostly AI-generated [7]. The difficulties of topic selection and coherent long-form content generation have led scammers to the low-hanging fruit of using AI to sufficiently re-write existing books to evade plagiarism filters [8].

In our work, we address the problem of finding prominent gaps in the literature and generating proposals for high-quality books that can fill them. Although the current marketplace is unfortunately being inundated with low-quality AI "slop", AI-written books do potentially offer real advantages: They can be easily kept up to date, whereas a traditional textbook on a rapidly evolving topic might be largely obsolete by the time it is written, edited, and published. AI-authored books can also target more niche topics than would attract human authors, and tune the curriculum for niche audiences and even individual readers. However, they must be targeted to fill the right gaps, where there is an honest need for them.

a) Contributions: Our main contributions include:

- *A method for detecting gaps in the literature:* We define a gap as a region in document embedding space that is filled with articles but devoid of books, and we present an efficient algorithm to identify these gaps. We apply Locality Preserving Projections (LPP) [9] to project books and articles into a shared subspace that localizes citation links. For 73 of 76 topics within the AI literature, our detected gaps are larger than every gap filled by an existing book on that topic.
- *Validation of detected gaps:* We validate our methodology along three axes:
  Feasibility of gap filling: LM-generated book proposals are, on median, closer to 96.4% of in-gap articles than any existing book.
  Historical justification: From 2000 to 2025, newly published books filled larger gaps on average than existing books (2.28 vs 1.61 articles), corroborating our definition and size-based selection of gaps.
  Human evaluation: In a blind survey of 22 participants, our LM-generated proposals were statistically indistinguishable (overlapping 95% confidence intervals) from published books on the same topics across all three criteria (interesting, well-written, publishable).

b) Organization: We begin by surveying related work in Section II. The OpenAlex research index of books and articles, and our process of deriving content-based embeddings of the works, is discussed in Section III. Geometric issues in defining and identifying gaps in the literature are discussed in Section IV. We present an approach to using an LM to fashion the title and abstract of a book proposal to fill an identified gap in Section V. We present our experimental results in identifying and filling gaps in 77 distinct topics in AI in Section VI. Finally, directions for building on our results are proposed in Section VII.

Source code and a reproducibility guide are available at https://github.com/GeoffChurch/bookgap.

Appendices are available at https://geoffchurch.github.io/bookgap/full.pdf.

II. Related Work

Our work bridges knowledge representation, scientometrics, and long-form text generation to identify and fill book gaps in the academic literature.

a) Embeddings: To properly identify gaps, we must first represent books and articles in a shared semantic space. Our approach uses text embeddings derived from titles and abstracts. Methods for text-based representation learning range from simple statistical baselines to heavy neural embedders. Although recent work tends to rely on heavy embedders such as SentenceBERT [10], simple representations like TF-IDF remain competitive for the scientometric tasks of predicting citation impact [11] and future research trends [12].

However, textual semantics do not necessarily capture useful signals embedded in the citation graph. To capture this structural information, Network Representation Learning (NRL) methods like DeepWalk [13] and Node2Vec [14] determine vector representations for nodes in a network. Graph Neural Networks (GNNs) [15] are designed to automatically leverage both content-based features (e.g. text embeddings) and network topology.

b) Prerequisite Mining: A textbook requires a pedagogical ordering. Curriculum mining extracts prerequisite relationships from, e.g., Wikipedia [16], course data [17], or directly from text [18]. We propose leveraging the citation graph as a proxy for dependency, as it in some sense captures the "self-education" of the research community.

c) Domain Adaptation: Using off-the-shelf content embedding models ignores the domain shift between document types. Books and articles have distinct stylistic conventions, which can create misalignment in the embedding space, triggering the detection of "false gaps". While unsupervised domain adaptation typically employs adversarial training [19] or instance weighting [20], we adopt a purely linear approach. We favor linear projections for their data efficiency, robustness, and strong *inductivity*: the learned transformation can project other work types, like book chapters, even if they are absent from the training data. We optimize for a projection that conflates class means, preserves within-class variance, and respects citations.

d) Gap Discovery: Gap mining is related to Literature-Based Discovery (LBD) [21], which traditionally identifies missing links in conceptual chains. This is often categorized into *closed discovery* (finding a bridge between two known concepts) and *open discovery* (generating new hypotheses from a starting concept) [22]. Modern LBD uses graph-based link prediction [23] or LLM-driven hypothesis generation [24]. Essentially, LBD seeks gaps in the published research within the space of possible research, whereas we seek gaps in the published books within the space of published articles.

e) Content Generation: Generating a textbook requires distilling vast amounts of source material into a long, structured narrative. A good textbook gives an intuitive, coherent picture of the topics it covers (a challenge of *abstraction*), while simultaneously grounding its coverage with direct references to trusted sources (a challenge of *extraction*). Generation tasks can furthermore be roughly categorized in terms of the sizes of the input and output, e.g. short-to-long or long-to-long [25]. Recursive summarization techniques have been applied to summarize entire books (long-to-short) [26] and, more similarly to textbooks, generate literature surveys (long-to-long) [27]. One key difference between a literature survey and a textbook is that a textbook should progress sequentially through a curriculum of topics.

State-of-the-art long-form generation relies on hierarchical planning. Systems like [28] and [29] alternate between high-level outlining and low-level elaboration to maintain coherence. Grounding this generation requires Retrieval-Augmented Generation (RAG) [30], where the model retrieves and embeds relevant snippets from trusted sources.

STORM [31] is a tool that, given a topic, will research, structure, and generate a high-quality Wikipedia-like article, grounded in source references. This is highly complimentary to our own task of identifying topics that are missing a textbook. An end-to-end pipeline could find a topic with many articles, books, blogs, or social media posts, but no Wikipedia article, and then prompt STORM to write such an article.

III. Data Preparation

The first step in our book proposal generation pipeline is representing the corpus of papers and books in a shared semantic space. In this section, we first describe the AI subfield of the OpenAlex index, and the steps we take to clean and deduplicate it into a suitable subset of books and articles. Next, we describe the embedding process and the domain adaptation techniques we use to identify a subspace where books and articles are better aligned.

A. Dataset

Our dataset is a collection of 1.56 million articles and 17,600 books in the Artificial Intelligence subfield of the OpenAlex [32] index.

OpenAlex is the official replacement for the Microsoft Academic Graph, which was retired in 2022. OpenAlex is fully open-source, and developed by the nonprofit OurResearch. It comprises data on and relations between five basic entity types: works, authors, venues, institutions, and concepts. We make use of works (books and articles) and concepts (topics within the AI subfield).

OpenAlex provides additional per-work data, including authorship and venue information, but we use only the following subset: **id**, **display_name**, **type**, **publication_date**, **language**, **referenced_works**, **primary_topic.id**, and **abstract_inverted_index** (the abstract, but stored as an inverted index).

a) Data Cleaning: Our analysis is sensitive to data artifacts in the OpenAlex corpus, such as duplicate entries and malformed and partial entries. After syntactic filtering and syntactic + semantic deduplication, 1,583,103 articles and 14,605 books remain.

We begin with 1,698,538 articles and 16,316 books from OpenAlex's 6 Nov 2025 snapshot, restricted to works that

TABLE I
COMPARISON OF EMBEDDERS, SHOWING THE NUMBER OF PARAMETERS, EMBEDDING DIMENSIONS, CONTEXT WINDOW SIZE, AND AVERAGE MTEB SCORE [35].

Name	Params	Emb. Dim.	Context	MTEB
GIST	33M	384	512	44.21
Gemma	300M	768	2048	61.15

have abstracts. The *syntactic filter* stage removes works with titles shorter than four characters and with abstracts shorter than ten characters, as well as works with ill-formed abstracts (e.g. containing `"An abstract is not available"`, `"Search for other works by this author"`, or `"Russian Text Ignored"`). The *syntactic deduplication* stage maps each work to its title+abstract, lower-cased and with all non-alphanumeric characters removed (including removing whitespace) to detect collisions, retaining only the earliest publication. Finally, the *semantic deduplication* stage filters out works one-by-one, in order from oldest to newest, that have a neighbor that is (1) older than it, (2) within a cosine distance threshold of 0.005, and (3) not filtered out when it was originally under consideration.

B. Content Embeddings

We compute vector embeddings for each of the 1.6 million works in our dataset. We experimented with two different embedding models, GISTEmbed small (GIST) [33] and EmbeddingGemma (Gemma) [34], summarized in Table I. Both embedders output L2-normalized vectors.

Although we have citation graph data at our disposal, we use content-based embeddings, derived solely from a work's title and abstract, for the following reasons:

- Content embeddings avoid the "cold-start" problem of determining embeddings for new works that have never been cited, and more generally the bias of older works having more citations.
- Content embeddings mitigate recency bias. Even "one-way" collaborative embeddings, where each work is represented by the earlier works that it cites, exhibit recency bias. This bias creeps in because the citation graph has bidirectional temporal locality: works tend to cite and be cited by works published around the same time. We are interested in topics, regardless of when they were being researched.
- We can evaluate our ability to find and plausibly fill book gaps by embedding the titles and abstracts of proposed books, without the confound of generating a full book with citations.

That being said, we do leverage the citation graph in our domain adaptation process (Section III-B1), but only to identify a projection that is applied to the content embeddings.

1) Domain Adaptation: Articles and books must occupy the same semantic space to facilitate our analysis. Raw content embeddings might lead us detect what appears to be an impressively large book gap only because the embedder over-prioritizes the stylistic features that systematically differ between articles and books. Any literature gap we discover is only meaningful to the extent that the books and articles are semantically aligned.

We restrict our optimization of the embeddings to linear projections, offering two distinct advantages:

- Many linear optimization problems in this domain are convex or have analytical solutions (e.g. generalized eigen-decomposition). This guarantees uniqueness and efficiency.
- Linear projections are *inductive*, so the learned transformation matrix can be used to embed new works even from previously unseen domains without re-training. This contrasts with *transductive* models, such as manifold learning's t-SNE or Laplacian Eigenmaps, which learn embeddings only for the training set.

a) Alignment via Citation Locality: To better align books and articles, we optimize for *citation locality*: works that are close in the citation graph should be close in the embedding space. This objective promotes alignment, because books and articles cite each other despite stylistic differences. To build the citation graph, we initialize an adjacency matrix A where $A_{ij} = 1$ if i cites j, and 0 otherwise. We set diagonal elements A_{ii} to 1 (virtual self-reference) for regularization. We then normalize outgoing citations such that if a work cites N other works, each edge weight is $1/N$.

The actual optimization is done via *Locality Preserving Projections* (LPP) [9]: we solve the generalized eigenproblem $X_c^\top L X_c \mathbf{a} = \lambda X_c^\top D X_c \mathbf{a}$ and retain the smallest eigenpairs. Here, X_c is the centered embeddings matrix (one row per work), L is the Laplacian of the symmetrized citation graph, and D is the diagonal matrix with D_{kk} equal to the sum of the weights of the edges incident on the kth work. Intuitively, LPP discards the directions along which embeddings are roughest when following citation links, such as directions that systematically distinguish books from articles, as well as directions that capture irrelevant properties within a class.

We also considered explicitly class-conflating objectives (Anti-LDA and Anti-CSP), and fusions of these with LPP. Plain LPP performed competitively against all of them (Table II), so we use plain LPP throughout the main experiments.

IV. BOOK GAP IDENTIFICATION

A. Defining Gaps

We define a gap based on a work's *p-ball*: the set of articles that lie within a fraction p of the cosine distance to its nearest book. We define an article's gap to be its $\frac{3}{4}$-ball, rather than its 1-ball or $\frac{1}{2}$-ball. See Section VI-C for an experimental justification of our choice of $p = \frac{3}{4}$.

An alternative definition of a gap might be based on a work's *Voronoi cell*, the set of articles closer to it than to

TABLE II
DENOISER COMPARISON FOR BOTH EMBEDDERS. EACH DENOISER DROPS 10 DIMENSIONS FROM GIST AND 394 FROM GEMMA, RETAINING 374. NONE MEANS NO DENOISING, ALL DIMENSIONS RETAINED (384 FOR GIST, 768 FOR GEMMA). SEE APPENDIX B FOR DENOISER DEFINITIONS.

Denoiser	Target gap coverage %age		Proposal gap %ile	
	GIST	Gemma	GIST	Gemma
NONE	81.82	81.02	65.98	36.37
CCLIP($\alpha = 0$)	83.80	86.22	73.93	69.71
CCLIP($\alpha = 0.5$)	85.26	**87.70**	79.00	81.09
CCLIP($\alpha = 1$)	83.77	83.79	**79.26**	81.75
Anti-LDA	84.03	87.33	78.15	76.27
JAD	83.59	83.62	79.10	74.78
LPP	**85.97**	85.46	74.14	**85.89**

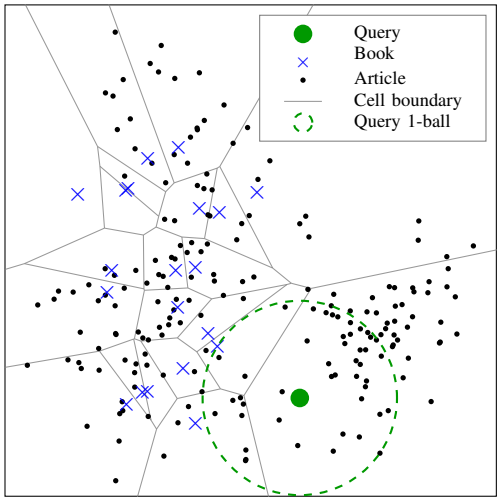


Fig. 1. A Voronoi diagram over a simulated set of 2D book embeddings and a query point, with the query's 1-ball which extends to the nearest book. Articles are scattered in the background.

any other book. See Fig. 1 for a visual comparison between p-balls and Voronoi cells. We define gaps using p-balls, rather than Voronoi cells, for two reasons:

- Voronoi cells are not geometrically centered around the point that defines them. Instead, their so-called *seed* or *generator* may be close to the cell wall, and the cell can easily span a huge portion of the semantic space. This is at odds with our intuitive conception of a gap as capturing a compact cluster of related works.
- Voronoi gaps are difficult to exploit in prompting an language model to generate good proposals (i.e. a title and abstract/outline pair). Proposals can be expected to land near the centroid of the works used to prompt the language model. This is convenient for a p-ball gap, where sampling from a uniform or center-preferring distribution will yield proposals that tend to fill that same ball. Voronoi gaps, on the other hand, are filled when the proposal's embedding lands near the seed of the cell, which can be very near to the edge of the cell. This creates a sharp discontinuity where the Voronoi cell of the generated proposal could go from containing thousands of articles to containing only a few, or even none at all. A good proposal would need to be very similar to articles that are close to the cell's generator/seed while being more dissimilar to articles outside of the cell, even when they fall within the same topic as the seed.

B. Data Structures for Finding Gaps

a) Algorithmic Gap Finding: To find the largest book gap among a set of books and articles, we compute the size of each article's gap and then pick the largest among those points. To avoid possible issues that could arise if trying to interpolate between articles, we restrict our search to points that represent actual articles.

To calculate the gap size for a point x, we find the distance $r = d(x, b)$ from it to the nearest book b, and then search for all articles a with $d(x, a) < r$. We use the Faiss library [36], specifically the in-memory Hierarchical Navigable Small World (HNSW) index, which offers both nearest-neighbor queries (for determining the distance r to the nearest book by querying an index built on books) as well as range queries (for determining the set of articles within distance r by querying an index built on articles).

V. FILLING GAPS

Once a book gap has been identified, we want to use generative model-based approaches to fill it. In this paper, we propose generating a title and abstract whose embedding will be located near the source of the identified gap. This requires transforming the contents of the gap into a coherent LM prompt that represents it.

a) Prompting the Language Model: Each proposed gap typically contains hundreds or thousands of articles, far too much content to fit within a single prompt. We therefore aggressively subsample the gap, selecting up to 10 in-gap articles uniformly at random, and picking up to 5 nearby books. Nearby books are selected by assigning each in-gap article its nearest book, and taking books in order of decreasing number of in-gap articles so covered. In our experiments, we prompt Gemma 4 (31B, instruction-tuned, Q4_K_M-quantized; Ollama tag `gemma4:31b-it-q4_K_M`) with appropriate books and articles to generate the proposed title and abstract (Appendix A). We include both books and articles, so that the LM is informed by (1) the stylistic norms of related books, which it can adopt, and (2) the focus of related books, so that it can avoid merely duplicating the focus of an existing book.

VI. EXPERIMENTAL RESULTS

In this section, we report results evaluating the performance of our gap finding/filling pipeline, through an end-to-end test of how well we can find and cover the largest gap in each of 77 topics in the AI literature.

In our experiments, we distinguish between the *target gap* and the *proposal gap*. The target gap is the (article-centered) gap we want to fill, and is used to prompt the LM for a

TABLE III
COMPARISON OF TARGET GAP COVERAGE PERCENTAGE AND PROPOSAL GAP PERCENTILE AVERAGED OVER THE 76 AI TOPICS THAT HAVE BOOKS, FOR VARIOUS SETTINGS OF THE NUMBER OF EXAMPLE ARTICLES AND BOOKS GIVEN TO THE LM, COMPARING GEMMA AND GIST.

References used		Target gap coverage %age		Proposal gap %ile	
art	bks	GIST	Gemma	GIST	Gemma
0	1	32.14	27.14	7.74	8.64
1	1	75.61	65.03	38.00	46.33
0	5	41.12	35.30	18.74	16.35
5	0	84.16	83.03	77.33	79.34
5	5	85.32	82.69	71.97	79.46
10	5	85.97	85.46	74.14	85.89

proposal. The proposal gap is the (proposal-centered) gap we actually filled, based on the embedding of the generated proposal. We expect a proposal gap to usually be smaller than the target gap that prompted it, but this is not necessarily the case as a proposal can land in an article-free zone that lets it cover more total articles. The metrics we report are:

- *target gap coverage percentage*: the percentage of the target gap articles that are closer to the proposal than to any existing book. This measures how well we were able to hit the intended target. Note that this is a Voronoi-like metric, rather than a p-ball-native metric like the percentage of the target gap articles that are also in the proposal gap. This is so that we get bigger (more robust) numbers and can in principle compare fairly across different gap definitions.
- *proposal gap percentile*: the percentile of the proposal gap's population (the number of articles in the gap) with respect to the empirical distribution of the gap populations of existing books. This measures how big of a gap we ended up filling, relative to existing books in that topic.

a) Methodology details: The numbers in Section VI are computed from GIST+LPP+Gemma 4 runs on the 77 AI topics. In experiments involving the Gemma embedder (Section VI-D), we drop 394 of Gemma's 768 dimensions to match GIST's retained 374D (dropping 10 of 384). The human evaluation in Section VI-F was conducted using proposals generated earlier with GPT-5 instead of Gemma 4.

A. Overall Performance

The generated proposals generally do a good job of hitting the intended target and achieving a high proposal gap percentile, provided enough context in the prompt. Table III shows that 10 in-gap articles and 5 nearby books achieve target gap coverage percentages of over 85% with both embedders, and proposal gap percentiles of over 74% with GIST and over 85% with Gemma.

Appendix E gives a per-topic breakdown of the system's performance, showing good performance on at least one metric for 69 of 77 topics. However, eight topics failed in both metrics. In each case, the proposal was as on-target as could be expected given the context. The failure modes are ascribable to some language dimensions surviving the denoiser, along with our prompt being in English (Appendices F-B to F-D and F-H to F-J), to data-quality issues that could be resolved by improved filtering (Appendices F-E and F-F), and to works that are misassigned by OpenAlex's classifier (Appendices F-C and F-E).

TABLE IV
PROPOSAL GAP PERCENTILE FOR THE RANDOM-CONTEXT BASELINE VS OUR GAP-BASED PIPELINE. THE PIPELINE BEATS THE BASELINE ON 45 TOPICS, LOSES ON 18, AND TIES ON 13. 10 OF THE 13 TIES OCCUR AT THE 0TH OR 100TH PERCENTILE. THE TWO-SIDED SIGN TEST ON THE 63 NON-TIED TOPICS GIVES $p = 0.00090$.

	Proposal gap percentile (%)	
	mean	median
Random baseline	58.8	74.2
Ours (gap-aware)	74.1	95.8

To validate that our pipeline finds and fills larger gaps than the proposal model would find on its own, we ran an ablation test (Table IV) in which the model was prompted to generate a gap-filling textbook based on 10 articles and 5 books randomly sampled from the topic, as well as the name of the topic to help bound the scope. The baseline does fill meaningfully large gaps (median of 74th percentile), but the pipeline proposals fill significantly larger gaps (median 95th percentile, $p < 10^{-3}$). This is to be expected, because the proposal model scored well on target gap coverage, meaning it can hit the target it is pointed to, and in the pipeline it is pointed to the largest gap by construction. So, the evaluation metric of proposal gap percentile trivially favors the pipeline, but the human evaluation in Section VI-F and historical evaluation in Section VI-E suggest that it does correlate with more grounded measures of proposal quality.

B. Sensitivity to Context

We provide a book gap to Gemma 4 with a prompt that contains a representative set of the titles and abstracts of in-gap articles and nearby books. In this section, we consider the effect of context volume on our gap coverage metrics.

Table III presents the impact of modulating the context, as specified by the number of books (0 to 5) and articles (0 to 10) included in the prompt. Article context proves to be far more valuable than book context. Including the single highest coverage book with no articles suffices to achieve target gap coverage percentages as high as 32.14%. This indicates that the proposal generation model is at least able to land near the provided book. It also suggests that, on average, each topic already has a book such that a good fraction of the articles in the largest gap are in its Voronoi cell. Articles provide far larger boosts to both metrics, especially to proposal gap percentile, which with one book reaches only $\approx 8\%$ (GIST) or $\approx 9\%$ (Gemma), but jumps to $\approx 38\%$ (GIST) or $\approx 46\%$ (Gemma) when we add an article.

TABLE V

p SWEEP. "COV" IS TARGET GAP COVERAGE PERCENTAGE, "PCT" IS PROPOSAL GAP PERCENTILE. THE MEAN AND MEDIAN OF EACH METRIC ARE COMPUTED OVER THE 77 AI TOPICS. "GAP MED" IS THE MEDIAN *target* GAP SIZE.

p	cov. mean	cov. med	pct. mean	pct. med	gap med
1/2	81.2	**100.0**	56.3	94.6	80
9/16	81.7	**100.0**	66.0	**98.4**	120
3/4	**86.2**	96.4	**74.1**	95.8	350
1	53.2	55.3	59.3	66.2	1016

C. Sensitivity to Radius Fraction

The p-ball definition of a gap (Section IV-A) defines the gap's radius as a fraction p of the cosine distance to the nearest book. We use $p = \frac{3}{4}$ throughout. To validate this choice, we sweep $p \in \{\frac{1}{2}, \frac{9}{16}, \frac{3}{4}, 1\}$, with the same LPP-denoised GIST embeddings used throughout (Table V).

We can observe the tension between size and specificity. When $p = 1$, the gaps are very large, but metric scores are poor due to contention with neighboring books. When $p < \frac{3}{4}$, the gaps are small, but the proposals attain 100% coverage on the majority of topics. $p = \frac{3}{4}$ is competitive in all cells, and leads decisively on the *average* target gap coverage percentage and proposal gap percentile. We choose $p = \frac{3}{4}$ as our default because it maximizes the means of both metrics while being an interior point of the parameter space, suggesting that smaller and larger p degenerate in one way or another.

D. Comparison of Embedding Models

The two embedders we investigated, GIST and Gemma, both perform well once denoised. With LPP, GIST's target gap coverage is higher in every row of Table III, while Gemma's proposal gap percentile is higher at the default (10 articles + 5 books), where neither metric differs significantly (GIST − Gemma is $+3.9 \pm 10.6$ on coverage and -12.9 ± 21.6 on percentile with 95% CIs). Without denoising, Gemma's proposals fill much smaller gaps than GIST's (proposal gap percentile is 36.37% vs 65.98%, Table II), so Gemma needs the denoiser whereas GIST only marginally benefits. This is partly a consequence of the experiment design, since matching 374 retained dimensions requires dropping 394 of Gemma's but only 10 of GIST's. However, Gemma does capture more of the book/article difference: a linear classifier trained on every book paired with a random article (without replacement) from the same topic had a mean within-topic AUC of 0.94 on Gemma's original embeddings but only 0.86 on GIST's (0.83 and 0.80 after LPP), and across ten repeats of the experiment (redrawing the paired articles and cross-validation folds), Gemma's AUC was higher in 68 to all 71 topics with ≥ 10 books.

E. Historical Evaluation of Books

Our primary objective here has been to find large empty gaps where research papers exist but books do not. One may question whether such criteria reflect the behavior of human authors. The historical record enables us to evaluate this hypothesis. We can look at the books actually published in a given year to see whether they filled unusually large gaps based on the literature to that date.

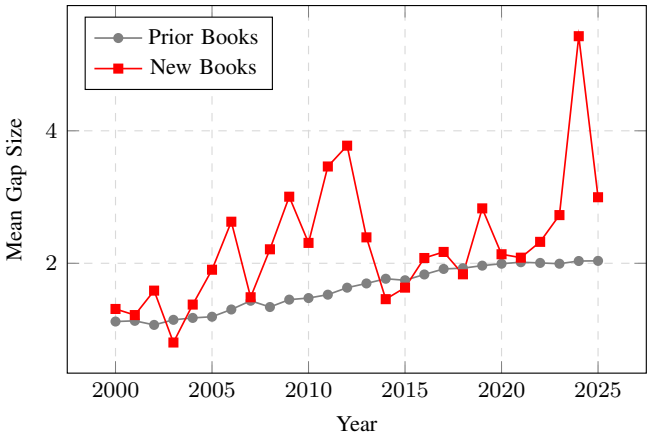


Fig. 2. For each year from 2000 to 2025, the mean gap size of prior books (published *before* that year) vs new books (published *in* that year), among articles and books published before that year.

Fig. 2 shows that new books do tend to fill larger gaps than existing books: from 2000 to 2025, newly published books covered larger gaps on average than previously published books (2.28 vs 1.61 articles). This helps justify our p-ball definition and size-based selection of gaps, and suggests that books populate new topics that have filled up with recent articles.

F. Subjective Evaluation

It is fair to question whether our proposals are truly interesting to people, as opposed to satisfying purely technical evaluation criteria. For this reason, we performed a subjective trial, asking 22 participants (five graduate students in AI and a Qualtrics-sourced pool of 17 participants with a master's degree or higher) to evaluate the quality of our proposed books against representative titles+abstracts of existing books. The survey instructions are reproduced in Appendix D.

For each of eight representative topics, we selected a highly-cited book, a less-cited book, and a book proposal we generated from the largest gap detected in that topic. Well-cited means the book was near the 95th percentile in citations among books with ≥ 2 citations published around the same time, and less-cited means it sat near the 30th percentile. We ruled out books whose abstracts mentioned years or authors, or included a table of contents. This was done so that the real books were not easily distinguishable from those of the proposals.

Table VI reports the mean rating per (book, criterion) cell, with 95% confidence intervals on the averages. Table VII gives the per-topic breakdown. Confidence intervals for all six difference pairs (`ours` vs `more`, `ours` vs `less` times the three criteria) include zero, so we do not rule out the null hypothesis that the three classes are pairwise indistinguishable.

TABLE VI

RESULTS OF HUMAN EVALUATION OF REAL AND GENERATED BOOK TITLES+ABSTRACTS ON THREE CRITERIA ON A 1-7 SCALE (HIGHER IS BETTER). IN EACH OF 8 TOPICS, WE PICKED ONE HIGHLY-CITED BOOK (`more`), ONE LESS-CITED BOOK (`less`), AND ONE GENERATED BOOK PROPOSAL (`ours`) FROM THE LARGEST GAP WE DETECTED IN THAT TOPIC. EACH CELL AVERAGES OVER THE RESPONDENTS WHO ANSWERED THE CORRESPONDING QUESTION, AND THEN OVER THE 8 TOPICS. THE $\pm$ NUMBERS ARE 95% CIS ACROSS THE 8 TOPICS.

	Interesting?	Well-written?	Publishable?
`more`	4.10 ± 0.20	4.34 ± 0.30	4.49 ± 0.25
`less`	4.14 ± 0.28	4.26 ± 0.26	4.36 ± 0.31
`ours`	4.14 ± 0.21	4.42 ± 0.21	4.43 ± 0.18

VII. CONCLUSIONS

We have demonstrated a pipeline for finding topics in need of books, and then generating proposals that would fill those gaps. The system performs well on synthetic metrics, and proves well grounded by historical analysis and human evaluation.

The obvious next direction is to actually generate full books to fill the holes we identified, rather than just proposing titles and abstracts. For this, we plan to leverage the citation graph to help build detailed, high-quality curricula that can ideally be used to guide existing tools like STORM [31] for the final task of generation. An end-to-end textbook generation pipeline would also enable more stringent (and expensive) evaluation of gap filling, as we could generate a book to target a given gap, and then evaluate a title and abstract derived solely from its table of contents and/or body text.

LIMITATIONS

This paper presents a rather speculative approach to identifying significant holes in the literature and attempting to fill them. It is best seen as a proof of concept rather than a production system, although we have evaluated its performance over 77 topics in the AI literature, using a mix of statistical, historical, and survey-based evaluation. Limitations of our work include:

- We have not yet evaluated our approach on topics outside those of the AI literature, because of the computational cost of the experiments.
- We did not include book chapters among our works, only articles and whole books. It would be helpful to know how many book chapters have covered a proposed topic, to distinguish between gaps that are completely uncovered vs those that are at least covered by chapters.
- We did not experiment with different approaches to sampling the articles that go into our LM prompt. We currently sample a fixed number of articles in a gap using a uniform distribution over the articles. Other possibilities are to sample based on citation popularity, or to sample more uniformly over the *space* covered by the gap, to increase topical diversity.

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APPENDIX A
EXAMPLE PROMPT FOR PROPOSAL GENERATION

We used the following prompt template (here filled in for the Natural Language Processing Techniques topic) to generate book proposals. See Appendix F-G for the generated proposal.

```
[system]
You are an academic researcher and educator.

[user]
Based on the research articles and related books
provided below, please propose a title and abstract for
 a textbook that would prepare students to understand
research in this area.

## Sample Articles

**Article 1**: Learning case-based knowledge for
disambiguating Chinese word segmentation
        *Abstract*: [...]

[Articles 2-10]

## Related Existing Books

**Book 1**: Joint Word Segmentation and Part-of-Speech
(POS) Tagging for Myanmar Language
        *Abstract*: [...]

[Books 2-5]

Output ONLY a JSON object with this exact schema, and
nothing else:
{"title": "...", "abstract": "..."}
No markdown fences, no commentary, no leading or
trailing text.
```

Fig. 3. Prompt for proposal generation.

APPENDIX B
PERFORMANCE COMPARISON OF DENOISING STRATEGIES

Table II reports the performance of the GIST and Gemma embedders when dropping k dimensions under several denoising strategies ($k = 10$ for GIST and $k = 394$ for Gemma, so both retain a 374D subspace). This experiment uses the same 77 AI topics used throughout the main text. The denoisers compared are: the $k = 0$ baseline "NONE" of no denoising; LPP (the default denoiser); Anti-LDA; joint approximate diagonalization (JAD) of the total S_T, Laplacian S_L, and per-class scatter matrices, each normalized to the same trace; and our *Class-Conflating Locality-Inducing Projections* (CCLIP), a convex combination of Anti-LDA's class-conflation and LPP's locality-induction numerators, but with S_T as the denominator.

We also considered explicitly class-conflating objectives:

- **Anti-LDA:** Anti-LDA is the reciprocal of Linear Discriminant Analysis (LDA) [37]. Instead of retaining the vector that maximizes the ratio of between-class variance to within-class variances, we suppress it. Specifically, we solve $S_W\mathbf{a} = \lambda S_B\mathbf{a}$ and retain the smallest eigenpairs. Here, S_W is the within-class scatter matrix and S_B is the between-class scatter matrix.
- **Anti-CSP:** Anti-CSP is the reciprocal of Common Spatial Patterns (CSP) [38]. Instead of retaining the directions that maximize the geometric difference between within-class variances, we suppress them. Specifically, we solve $S_{\text{art}}\mathbf{a} = \lambda S_{\text{book}}\mathbf{a}$ and retain the eigenpairs with the smallest $|\log\lambda|$, i.e. where the ratio of class variances is (geometrically) closest to 1.
- **Fused objectives:** Fusions of the above objectives, such as combining Anti-LDA and LPP into a single generalized eigenproblem (GEP), or joint approximate diagonalization (JAD) [39] of various combinations of the Laplacian, total scatter, and within-class scatter matrices.

Plain LPP performed competitively across our evaluation metrics (*target gap coverage percentage* and *proposal gap percentile*, see Section VI). This suggests that promoting citation locality promotes class conflation as a side effect: LPP makes it harder for a linear classifier to distinguish books and articles (Section VI-D), and with Gemma embeddings it raises the proposal gap percentile (Table II) over NONE by $+49.5 \pm 10.8$ (95% CI), while the explicitly class-conflating Anti-LDA attains a more modest $+39.9 \pm 11.6$ (LPP $-$ Anti-LDA is $+9.6 \pm 8.5$). However, the space of possible objectives is large and not systematically explored here.

a) Applying Denoising: All our denoising strategies are based on identifying a certain subspace. The eigenvectors P returned by GEP- or JAD-based denoisers are not necessarily orthogonal, so using them directly to project our embeddings can squash the embeddings into the right subspace but with undesirable skew. We therefore orthonormalize them using a thin QR decomposition to discard R and retain only Q. Finally, after projecting the embeddings with the (thin) Q we re-normalize them back to unit length.

CCLIP solves the GEP $\left((1-\alpha)\,\tilde{S}_B + \alpha\,\tilde{S}_L\right)\,\mathbf{a} = \lambda\,S_T\,\mathbf{a}$ and retains the smallest eigenpairs, where $\tilde{S}_B$ and $\tilde{S}_L$ are the between-class scatter S_B and citation graph Laplacian scatter S_L, but normalized to have trace equal to the original embedding dimension (384 for GIST, 768 for Gemma). Trace normalization ensures that $\alpha = 0.5$ fairly balances the two objectives (unnormalized $\mathrm{tr}(S_L)$ is much larger than $\mathrm{tr}(S_B)$ on our embeddings). The numerator interpolates between class conflation ($\alpha = 0$) and locality induction ($\alpha = 1$), but the endpoints do not exactly recover Anti-LDA or LPP since their denominators are within-class scatter S_W and degree-weighted scatter $X_c^\top D X_c$, respectively, rather than S_T.

Every denoiser significantly improves Gemma's proposal gap percentile by at least 33 percentage points over NONE on average. LPP's gains over NONE are not quite significant elsewhere (Fig. 4); in general, GIST benefits only marginally from denoising (Section VI-D). Since no denoiser is significantly better than LPP on either embedder or metric, we adopt LPP throughout the main text for its simplicity and because it is the optimal linear (hence inductive) solution of the Laplacian Eigenmaps problem [9].

The detailed, per-topic results for the default GIST run (dropping 10 dimensions with LPP, and passing 10 articles and

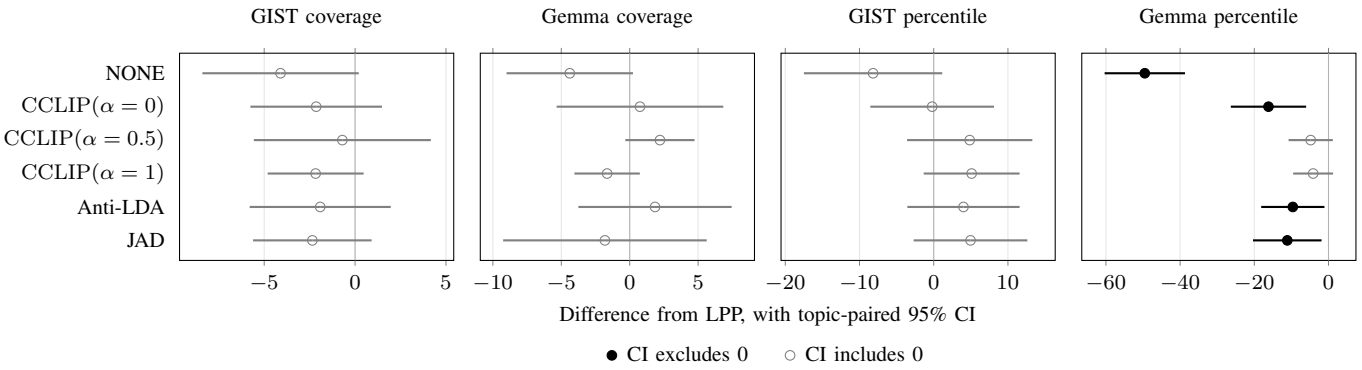


Fig. 4. Each denoiser's mean difference from LPP over the 77 topics (76 for proposal gap percentile, as T13567 has no books and has in fact been emptied entirely since our data snapshot was taken), with 95% topic-paired CIs. The only significant differences at 95% confidence are LPP's four wins on Gemma's proposal gap percentile. Every denoiser (except NONE, which leaves the embeddings untouched) retains 374 dimensions by dropping 10 from GIST and 394 from Gemma.

5 books to the LM) are shown in Table VIII. The target gap percentile (the percentile of the detected gap size with respect to the empirical distribution of actual book gap sizes) gives us an indication, without considering the generated proposals, of the performance of our gap detection algorithm (of course it is ultimately irrelevant if we cannot fill the target gap). This percentile is almost always 100%, so gap detection is working as intended.

APPENDIX C
PER-TOPIC HUMAN EVALUATION RESULTS

TABLE VII

RESULTS OF HUMAN EVALUATION OF REAL AND GENERATED BOOK TITLES+ABSTRACTS ON THREE CRITERIA ON A 1-7 SCALE (HIGHER IS BETTER). IN EACH OF 8 TOPICS, WE PICKED ONE HIGHLY-CITED BOOK (MORE), ONE LESS-CITED BOOK (LESS), AND ONE GENERATED BOOK PROPOSAL (OURS) FROM THE LARGEST GAP WE DETECTED IN THAT TOPIC. EACH CELL AVERAGES OVER THE RESPONDENTS WHO ANSWERED THE CORRESPONDING QUESTION. THE BOTTOM ROWS SHOW THE AVERAGES AND 95% CONFIDENCE INTERVALS ACROSS THE 8 TOPICS.

Topic	Interesting?			Well-written?			Publishable?		
	more	less	ours	more	less	ours	more	less	ours
Natural Language Processing Techn›	4.14	3.50	**4.36**	4.36	3.68	**4.64**	**4.73**	3.73	4.68
Semantic Web and Ontologies	**4.00**	3.95	3.59	**4.59**	4.23	4.05	**4.50**	4.32	4.09
Neural Networks and Applications	**4.24**	4.18	4.13	**4.86**	4.73	4.52	**4.73**	**4.73**	4.22
Fuzzy Logic and Control Systems	4.23	**4.32**	4.00	4.14	**4.41**	4.36	4.45	**4.50**	4.36
Algorithms and Data Compression	3.77	4.00	**4.18**	4.09	4.14	**4.45**	4.23	4.36	**4.45**
Artificial Intelligence in Games	4.50	**4.65**	4.27	**4.77**	4.48	4.64	**4.95**	4.87	4.73
Intelligent Tutoring Systems and ›	3.81	4.22	**4.27**	3.95	**4.26**	4.05	4.09	4.39	**4.41**
Cognitive Science and Mapping	4.14	4.27	**4.33**	3.95	4.14	**4.68**	4.27	4.00	**4.50**
Average	4.10	4.14	4.14	4.34	4.26	4.42	4.49	4.36	4.43
CI	0.20	0.28	0.21	0.30	0.26	0.21	0.25	0.31	0.18

APPENDIX D
SURVEY INSTRUCTIONS

This appendix contains the instructions shown to survey participants. The order of the 24 books was randomized for each participant. Participants did not know which were human-written vs LM-generated. They did know from the instructions that some were LM-generated, and that the purpose of the survey was to evaluate the quality of the proposal pipeline.

Welcome page

Survey: Textbook Proposal Evaluation

Goal: This survey aims to evaluate the performance of an end-to-end system for proposing new textbooks, against human-authored proposals.

What to expect: You will be presented with a randomized list of proposals (titles and abstracts). Some are generated by our automated system and others are written by humans. You will be asked to evaluate each proposal based on:

Topic Selection Is the book interesting to you?
Writing Quality Is the proposal is well-written?
Publishing Value Is it worth writing and publishing the corresponding book?

Estimated time: 20 minutes.

Thank you for participating in this study!

Demographic questions

Who are you? (single-choice)

- undergraduate student
- graduate student
- faculty member
- industry professional (text entry: which industry?)
- other (text entry)

How knowledgeable are you about the following?

- Artificial intelligence [7-point scale]
- Academic/textbook publishing [7-point scale]

Per-book question template

For each of the 24 books, participants saw the following:

Title: ⟨*book title*⟩
Abstract: ⟨*book abstract*⟩

Do you personally think this book is...

- interesting? [7-point scale]
- well-written? [7-point scale]
- worth publishing? [7-point scale]

APPENDIX E
PER-TOPIC BREAKDOWN OF EXPERIMENT METRICS

TABLE VIII
PER-TOPIC RESULTS FOR GIST WITH DENOISER LPP AT $k = 10$.

Topic	books	articles	Target gap coverage %age	Proposal gap %ile	Target gap %ile
Quantum Information and Cryptography	44	46542	94.0	95.5	100.0
Topic Modeling	58	38679	99.6	100.0	100.0
Metaheuristic Optimization Algorithms...	115	27451	93.8	97.4	100.0
Logic, programming, and type systems	338	26107	100.0	100.0	100.0
Natural Language Processing Techniques	1231	87408	99.2	100.0	100.0
Speech Recognition and Synthesis	110	33860	97.6	98.2	100.0
Semantic Web and Ontologies	620	52487	6.1	0.0	100.0
Cryptography and Data Security	47	28375	97.7	95.7	100.0
Neural Networks and Applications	817	73798	89.5	95.1	100.0
Multi-Agent Systems and Negotiation	166	16979	90.9	100.0	100.0
Reinforcement Learning in Robotics	42	12590	97.6	95.2	100.0
Advanced Clustering Algorithms Research	39	10374	96.0	100.0	100.0
Advanced Software Engineering Methodo...	171	26257	0.0	0.0	100.0
Sentiment Analysis and Opinion Mining	30	19750	99.2	0.0	100.0
Quantum Computing Algorithms and Arch...	197	22648	89.7	95.9	100.0
Target Tracking and Data Fusion in Se...	68	32001	76.8	0.0	100.0
Privacy-Preserving Technologies in Data	63	21623	98.5	95.2	100.0
Fuzzy Logic and Control Systems	228	21484	99.6	92.5	100.0
AI in cancer detection	57	19494	96.3	98.2	100.0
AI-based Problem Solving and Planning	171	16414	99.5	100.0	100.0
Cryptographic Implementations and Sec...	54	13731	98.2	98.1	100.0
Logic, Reasoning, and Knowledge	403	18712	90.7	100.0	100.0
Coding theory and cryptography	135	21971	96.2	0.0	100.0
Algorithms and Data Compression	182	22630	87.1	87.9	100.0
Advanced Graph Neural Networks	20	8822	99.0	100.0	100.0
Solar Radiation and Photovoltaics	37	15229	98.9	0.0	100.0
Bayesian Modeling and Causal Inference	141	13949	91.3	0.0	100.0
Domain Adaptation and Few-Shot Learning	9	8866	96.6	100.0	100.0
Security and Verification in Computing	52	13979	95.9	98.1	100.0
Anomaly Detection Techniques and Appl...	84	21075	98.5	100.0	100.0
Text and Document Classification Tech...	30	10771	98.8	96.7	100.0
Artificial Intelligence in Games	237	10467	95.5	100.0	100.0
Internet Traffic Analysis and Secure ...	76	13924	99.8	52.6	100.0
Stochastic Gradient Optimization Tech...	10	2763	97.2	100.0	100.0
Imbalanced Data Classification Techni...	33	9605	95.8	100.0	100.0
Adversarial Robustness in Machine Lea...	17	9123	96.5	100.0	100.0
Bayesian Methods and Mixture Models	101	14915	92.5	80.2	100.0
Intelligent Tutoring Systems and Adap...	177	11891	98.6	100.0	100.0
Evolutionary Algorithms and Applications	151	16648	97.0	94.7	100.0

TABLE VIII
PER-TOPIC RESULTS FOR GIST WITH DENOISER LPP AT $k = 10$. (CONTINUED)

Topic	books	articles	Target gap coverage %age	Proposal gap %ile	Target gap %ile
Explainable Artificial Intelligence (...	47	4908	100.0	100.0	100.0
Speech and dialogue systems	157	17839	98.1	93.0	100.0
Machine Learning and Algorithms	35	8397	92.4	94.3	97.1
AI in Service Interactions	78	9203	90.3	96.2	100.0
Wireless Signal Modulation Classifica...	13	6759	99.9	92.3	100.0
Geochemistry and Geologic Mapping	2125	211580	17.9	0.0	100.0
Educational Technology and Pedagogy	8	26968	99.9	100.0	100.0
Hate Speech and Cyberbullying Detection	187	10032	99.5	99.5	100.0
Authorship Attribution and Profiling	78	4271	82.1	88.5	100.0
Machine Learning and Data Classification	57	7700	96.2	100.0	100.0
Neural Networks and Reservoir Computing	14	7409	98.3	85.7	100.0
Machine Learning and ELM	1	4853	99.9	0.0	0.0
Data Stream Mining Techniques	25	6464	98.7	100.0	100.0
Cognitive Science and Mapping	459	8957	96.2	100.0	100.0
Gaussian Processes and Bayesian Infer...	31	4845	96.6	67.7	100.0
Seismology and Earthquake Studies	109	12347	96.4	100.0	100.0
Cognitive Computing and Networks	96	4288	97.7	81.2	100.0
Advanced Text Analysis Techniques	62	14382	94.3	95.2	100.0
Data Analysis with R	380	7570	99.7	99.2	100.0
Experience-Based Knowledge Management	6	606	89.3	16.7	83.3
Edcuational Technology Systems	105	40187	90.3	99.0	100.0
AI and Multimedia in Education	0	1153	100.0		
Linguistic Studies and Language Acqui...	731	21431	0.0	0.0	100.0
Text Readability and Simplification	30	3522	97.0	100.0	100.0
Computational Physics and Python Appl...	1988	19085	0.0	0.0	100.0
Computer Science and Engineering	48	18729	9.6	0.0	100.0
Machine Learning in Healthcare	31	5474	91.9	0.0	100.0
Advanced Computational Techniques and...	350	141656	87.4	72.9	100.0
Law, AI, and Intellectual Property	189	5812	86.6	0.0	100.0
Diverse Interdisciplinary Research St...	29	1771	99.0	100.0	100.0
Artificial Intelligence Applications	122	2394	92.9	100.0	100.0
Mathematical Control Systems and Anal...	30	3606	97.1	100.0	100.0
Organizational and Employee Performance	36	11372	0.0	0.0	100.0
Intuitionistic Fuzzy Systems Applicat...	20	1327	96.0	95.0	100.0
Educational Robotics and Engineering	164	21512	100.0	100.0	100.0
Statistical and Computational Modeling	57	4763	87.7	91.2	100.0
Psychiatry, Mental Health, Neuroscience	140	2321	12.7	0.0	100.0
Advanced Technologies in Various Fields	6	4218	96.6	100.0	100.0

APPENDIX F
PER-TOPIC GAPS, PROMPTS, AND PROPOSALS

This appendix details the gap, prompt, and proposal for each of ten selected topics from the GIST+LPP+Gemma 4 (10 articles, 5 books) run: the eight "failure" topics described in Section VI-A and two representative success cases (`T10181`, Natural Language Processing Techniques, and `T11273`, Advanced Graph Neural Networks). For each topic, we provide a visualization of the gap via PCA of the in-gap articles to 2D. By omitting the nearby books and the generated proposal from the PCA, the visualization gives a sense of how the books and proposal relate to the gap "on its own terms". In the visualization, the highest-coverage existing book has full size and opacity, both of which decay for lower-coverage books.

A. T11273 *Advanced Graph Neural Networks*

Topic has 20 books and 8822 articles; proposal covers 99.0% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Modern Graph Representation Learning: Theory, Algorithms, and Industrial Applications**

This textbook provides a comprehensive introduction to the field of graph representation learning, designed to equip students and researchers with the theoretical foundations and practical skills necessary to navigate modern research in network embedding. Starting with the fundamental concepts of node and edge embeddings, the text explores a wide array of architectures, including Graph Neural Networks (GNNs), Graph Attention Networks, and autoencoders. Special emphasis is placed on the nuances of different graph topologies, such as bipartite and heterogeneous networks, and the integration of auxiliary attribute information to enhance embedding quality. The book further bridges the gap between theory and practice by discussing the interplay between transductive and inductive learning, the challenges of network alignment across attributed networks, and the engineering requirements for scaling these models to web-scale industrial data. Through a combination of algorithmic analysis and real-world applications—including link prediction and node classification—this volume serves as a definitive guide for mastering the art of transforming complex graph structures into actionable latent representations.

b) Prompt articles:

1) W3080525460 **Graph Attention Networks over Edge Content-Based Channels**
2) W3193311899 **Integrating Transductive and Inductive Embeddings Improves Link Prediction...**
3) W4386768914 **EmbedX: A Versatile, Efficient and Scalable Platform to Embed Both Graphs...**
4) W2920698537 **Anchor Link Prediction across Attributed Networks via Network Embedding**
5) W3130159758 **PINE : Universal Deep Embedding for Graph Nodes via Partial Permutation...**
6) W3011167897 **Learning Vertex Representations for Bipartite Networks**
7) W3010668424 **Research on Bipartite Network Embedding with Auxiliary Information**
8) W3156855620 **BRIGHT: A Bridging Algorithm for Network Alignment**
9) W3108202858 **A Survey on Heterogeneous Graph Embedding: Methods, Techniques...**
10) W4283207431 **Decision-based Sampling for Node Context Representation**

c) Prompt books:

1) W3160730739 **Deep Learning on Graphs**
2) W4318815354 **Graph Neural Networks for Natural Language Processing: A Survey**
3) W3109650979 **Graph Kernels: State-of-the-Art and Future Challenges**
4) W4285722849 **Knowledge Graphs: An Information Retrieval Perspective**
5) W6930704475 **Book of Abstracts of France's International Conference on Complex Systems...**

B. T10639 Advanced Software Engineering Methodologies

Topic has 171 books and 26257 articles; proposal covers 0.0% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Software Architecture and Design: From Requirements Engineering to Product Line Evolution**

This textbook provides a comprehensive foundation in software architecture and design, bridging the gap between theoretical research and industrial application. It guides students through the end-to-end process of system creation, beginning with the elicitation of organizational goals and business modeling, and progressing through the analysis of functional and non-functional requirements. The text delves deeply into architectural synthesis, covering reference architectures, formal specifications, and the use of architectural patterns to manage system complexity. Furthermore, it introduces advanced paradigms for scalability and reuse, specifically focusing on Software Product Line (SPL) engineering and the strategic integration of legacy systems. By synthesizing model-driven development, aspect-oriented design, and pragmatic architectural principles, this book prepares students to design maintainable, extensible, and high-quality software systems in both small-scale and enterprise environments.

b) Prompt articles:

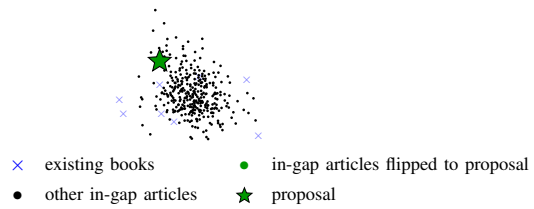
1) W27296737 **Diseñando Arquitecturas de Referencia para Integrar Sistemas Heredados...**
2) W7100737482 **considerando Requisitos No-Funcionales**
3) W2590145055 **Prototipado de programas usando especificaciones funcionales**
4) W293850605 **Elicitación de Objetivos, un estudio comparativo**
5) W805822484 **Diagramas de estructuras como mapa del sistema**
6) W1593945029 **Aspectos tempranos: Un enfoque basado en tarjetas CRC**
7) W2186467201 **La especificación formal en contexto: actual y futuro Formal specification...**
8) W1590387804 **Un modelo genérico para el modelo de negocio**
9) W331416522 **Un cálculo de patrones arquitectónicos**
10) W376743325 **Proceso de elicitación de objetivos**

c) Prompt books:

1) W4292210684 **Softwarearchitektur pragmatisch**
2) W6894286308 **Combined Modeling of Software and Hardware with Versions and Variants**
3) W1501652040 **Software Product Lines: Practices and Patterns**
4) W2132139515 **Software Architecture and Design: Principles, Models, and Methods**
5) W1554977156 **Design and Use of Software Architectures: Adopting and Evolving a...**

C. T13650 Computational Physics and Python Applications

Topic has 1988 books and 19085 articles; proposal covers 0.0% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Foundations of Interdisciplinary Research: Methods, Analysis, and Application**

This textbook provides a comprehensive introduction to the methodologies of academic research across diverse fields, including law, medicine, pedagogy, and the natural sciences. By analyzing a wide array of case studies—ranging from the legal frameworks of the social solidarity economy and human rights guarantees to clinical oncohematology and geological history—students will learn how to synthesize complex data, apply specialized diagnostic tools, and navigate the intersection of different academic disciplines. The text emphasizes the importance of semantic technologies in managing scientific literature and the role of globalization in shaping modern educational and social structures, equipping students with the critical thinking skills necessary to conduct rigorous research in an increasingly interconnected academic landscape.

b) Prompt articles:

1) W6926342650 **ПРАВОВЫЕ ТЕХНОЛОГИИ ОРГАНИЗАЦИИ ТРУДА В СОЦИАЛЬНОЙ СОЛИДАРНОЙ...**
2) W6926328260 **Правовой анализ законодательства некоторых зарубежных стран о...**
3) W6926491370 **СОЗДАНИЕ ИСКУССТВЕННЫХ ЗЕМЕЛЬНЫХ УЧАСТКОВ: МИРОВОЙ ОПЫТ**
4) W6926250510 **КРИМИНОЛОГИЧЕСКИЕ ОСОБЕННОСТИ ЛИЦ, СОВЕРШАЮЩИХ ПРЕСТУПЛЕНИЯ В...**
5) W6926297188 **ПЕДАГОГИЧЕСКАЯ ИНТЕНЦИОНАЛЬНОСТЬ КАК КОМПОНЕНТ НАПРАВЛЕННОСТИ...**
6) W6944971714 **Приоритетные направления развития транспортной инфраструтуры...**
7) W6926537574 **О ПРОБЛЕМЕ СОВРЕМЕННОГО ОБРАЗОВАНИЯ В УСЛОВИЯХ ГЛОБАЛИЗАЦИИ**
8) W6925493041 **Полибий о Понте Эвксинком: история геологического времени**
9) W6926456657 **ЧАСТОТА ВСТРЕЧАЕМОСТИ И КЛИНИЧЕСКАЯ КАРТИНА...**
10) W7099523390 **УДК 004.891**

c) Prompt books:

1) W4210359244 **Углубленный курс информатики**
2) W7069376991 **რჩეული თხზულებები**
3) W7038990522 **Le retour du tragique**
4) W6981720946 **Fachbereich Sozial- und Kulturwissenschaften**
5) W6987283722 **Speciální pedagogika předškolního a školního věku a dospělých: Distanční...**

D. T13674 Computer Science and Engineering

Topic has 48 books and 18729 articles; proposal covers 9.6% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Fundamentals of Digital Image Processing and Information Security**

This textbook provides a comprehensive introduction to the intersecting fields of digital image processing, pattern recognition, and data security. Designed for students entering the realm of computational research, the book covers essential topics including image enhancement, compression, and edge detection, as well as the mathematical foundations of cryptography, such as symmetric and asymmetric encryption algorithms and steganography. By bridging the gap between theoretical frameworks and real-world applications—ranging from automated vehicle classification and biometric recognition to secure file transmission—this text equips students with the necessary tools to analyze and implement advanced systems in computer vision and cybersecurity.

b) Prompt articles:

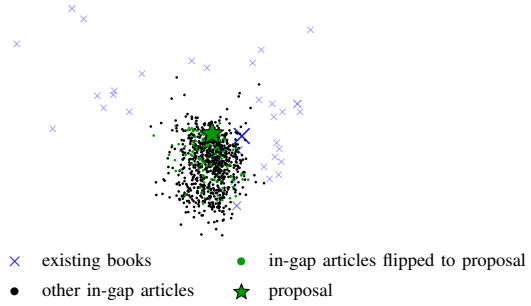
1) W2619042931 **Pengenalan Golongan Kendaraan Untuk Tarif Tol Berbasis Pengolahan Citra...**
2) W3102271506 **SISTEM PENDETEKSI KERUSAKAN LUAR ANGKUTAN UMUM**
3) W2749351705 **Sistem keamanan data metode kriptografi menggunakan algoritma EPA**
4) W3160903219 **Implementasi Metode Advance Image Coding UntukImage Compresion Pada Citra...**
5) W3017070376 **Analisis dan Implementasi Algoritma Rijndael (AES) dan Kriptografi RSA...**
6) W2607528082 **Pengenalan Ucapan Menggunakan Algoritma Back Propagation**
7) W2786622117 **DESAIN AES (ADVANCED ENCRYPTION STANDARD)**
8) W2910758262 **Metode Dynamic Parity Bit Steganografi Berdasarkan Modifikasi Metode Least...**
9) W2346006152 **PENGEMBANGAN SISTEM PEMETAAN KAWASAN BERBENTUK KORIDOR YANG BERBIAYA...**
10) W3116287321 **IMPLEMENTASI ENKRIPSI DALAM PENGAMANAN FILE DATA KARYAWAN DENGAN METODE...**

c) Prompt books:

1) W2468231853 **Information Hiding Text in Image: Encryption**
2) W2912959504 **Pengolahan Citra : Dasar dan Contoh Penerapannya**
3) W4385393770 **Transformasi PAI dalam Penguatan Sikap Moderasi Beragama di Era Digital**
4) W217728680 **Pemanfaatan Robot Sebagai Pendeteksi Korban Bencana Alam Menggunakan...**
5) W4319332415 **ALGORITMA HPCMA: REKAYASA ALGORITMA MEMETIKA UNTUK PENGENALAN SIDIKJARI**

E. T12157 Geochemistry and Geologic Mapping

Topic has 2125 books and 211580 articles; proposal covers 17.9% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Fundamentals of Systematic Zoology and Biogeography**

This textbook provides a comprehensive introduction to the principles of taxonomy, systematics, and biogeography, designed to prepare students for research in biodiversity documentation. Through an examination of species descriptions, distribution records, and taxonomic revisions, the text covers essential methodologies for identifying taxa, managing nomenclatural stability, and mapping zoogeographical distributions. By integrating case studies from entomology, arachnology, and mammalogy, students will develop the skills necessary to analyze morphological data, navigate museum records, and contribute to the global understanding of species occurrence and distribution patterns.

b) Prompt articles:

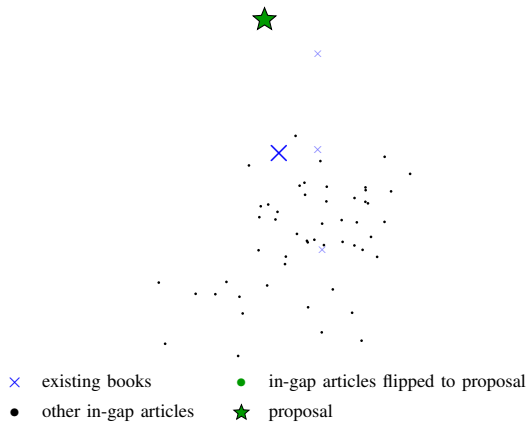
1) W7079589420 **Ksenoplata medicaginis Boucek 1965**
2) W7079704078 **Veerabahuthrips longicornis Okajima 1978**
3) W7079528891 **Stenchaetothrips caulis Bhatti 1982**
4) W7079544578 **Acallurothrips hagai Okajima 1993**
5) W7078072178 **Tegenaria lenkoranica**
6) W7078240407 **Diaea livens Simon 1876**
7) W7078239779 **Dysdera dunini Deeleman-Reinhold 1988**
8) W7078168628 **Trichothyse senilis**
9) W7078417866 **Tenuiphantes tenuis**
10) W7079462356 **Coloradoa viridis**

c) Prompt books:

1) W6949141153 **Bats from Polynesia, Melanesia, and Malaysia**
2) W2490043622 **Notice sur les gisements des lentilles trilobitifères taconiques de la...**
3) W7084074731 **A Review of the Old World Genus Camptorhinus SCHÖNHERR, 1825 (Coleoptera...**
4) W7080138678 **Qərbi Azərbaycan: I cild: Vedibasar mahalı**
5) W2960264760 **A Geological Reconnaissance of the Coal Fields of the Indian Territory**

F. T13623 *Linguistic Studies and Language Acquisition*

Topic has 731 books and 21431 articles; proposal covers 0.0% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Foundations of Italian Studies: Language, Literature, and Translation**

This comprehensive textbook introduces students to the interdisciplinary field of Italian studies, bridging the gap between linguistic theory, literary analysis, and cultural pedagogy. By examining the evolution of the Italian language, the nuances of translation between Italian and other global languages, and the study of both classical and contemporary Italian literature, this volume provides the essential tools for navigating academic research in the humanities. Students will explore the intersection of language acquisition, onomastics, and the socio-cultural dynamics of Italian identity, preparing them for advanced research in Romance languages and comparative literature.

b) Prompt articles:

1) W2799291577 **Marchione, Margherita. Clemente Rebora . Boston: G.K. Hall, 1979**
2) W2802159909 **Bernardo, Aldo S. and Rigo Mignani. Ritratto Dell'Italia. 2nd Ed...**
3) W2799542432 **Ciljberti, Anna and Giovanna Stefancich. Come leggere L'Italia d'oggi...**
4) W2800984227 **Pucciani, Oreste F. and Jacqueline Hamel. Langue et langage , 4° edition...**
5) W2799999873 **Carton, Dana and Anthony Caprio. En français: French for Communication...**
6) W2801874030 **Cancellotii, D. and C. Zanoni. Fabbricato in Italia . Cambridge: Cambridge...**
7) W2803040112 **Finco, Aldo. Appunti. Lubbock, Texas: Texas Tech University Bookstore...**
8) W2800500412 **Vellaccio, Lydia and M. C. Elston. Let's talk Italian . London: University...**
9) W2805876466 **Bartocci, Gianni. I Volsci a Dublino . Parma : Intelisano, 1968**
10) W2799646585 **Silone, Ignazio. Fontamara . Edited by C. Bartoli Perrault and M. J...**

c) Prompt books:

1) W2753435604 **Antologia Bilíngue: Clássicos da Língua Italiana**
2) W4299934308 **Dialoghi traduttologici: il testo letterario e la lingua inglese, Chieti...**
3) W595192858 **L'onomastica dell'Italia antica : aspetti linguistici, storici, culturali...**
4) W2753562188 **A literatura italiana no Brasil e a literatura brasileira na Itália: sob o...**

G. T10181 Natural Language Processing Techniques

Topic has 1231 books and 87408 articles; proposal covers 99.2% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Computational Chinese Word Segmentation: Theory, Methods, and Applications**

This textbook provides a comprehensive introduction to the fundamental task of Chinese Word Segmentation (CWS), a cornerstone of Chinese natural language processing. Beginning with the linguistic challenges of ambiguity and out-of-vocabulary words, the text traces the evolution of segmentation strategies from traditional dictionary-based and rule-based approaches to modern statistical machine learning and deep neural architectures. Students will explore key methodologies including sequence labeling, Conditional Random Fields (CRF), and attention-based neural networks, as well as the benefits of joint modeling for CWS and Part-of-Speech (POS) tagging. By examining the integration of unsupervised learning and the impact of segmentation on downstream applications such as machine translation and information retrieval, this book equips students with the theoretical foundation and practical knowledge necessary to engage with current research in the field.

b) Prompt articles:

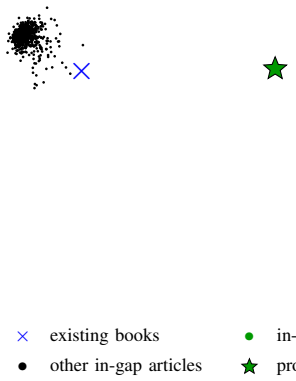
1) W1984676939 **Learning case-based knowledge for disambiguating Chinese word segmentation**
2) W2905474383 **Unsupervised Learning Helps Supervised Neural Word Segmentation**
3) W2379663637 **A Comparison Study of Sequence Labeling Methods for Chinese Word...**
4) W3116348602 **Joint Chinese Word Segmentation and Part-of-speech Tagging via...**
5) W3021494377 **Chinese Word Segmentation by Classification of Characters.**
6) W97157637 **Chinese Word Segmentation Using Various Dictionaries**
7) W2004827279 **Chinese base phrases chunking based on latent semi-CRF model**
8) W2035686871 **The Review of Automatic Chinese Word Segmentation**
9) W2913095682 **Sequence Labeling of Chinese Text Based on Bidirectional Gru-Cnn-Crf Model**
10) W2787109023 **Do We Need Chinese Word Segmentation for Statistical Machine Translation**

c) Prompt books:

1) W3026077554 **Joint Word Segmentation and Part-of-Speech (POS) Tagging for Myanmar...**
2) W2913914109 **Mandarin Chinese Words and Parts of Speech: A Corpus-based Study**
3) W4246718325 **Text Segmentation**
4) W3211848854 **Statistical Machine Translation**
5) W3139019561 **Deep Learning and Linguistic Representation**

H. T14064 Organizational and Employee Performance

Topic has 36 books and 11372 articles; proposal covers 0.0% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Organizational Dynamics and Institutional Development: A Comprehensive Guide to Leadership, Innovation, and Performance**

This textbook provides a multidisciplinary framework for understanding the complexities of modern organizational management and institutional growth. By synthesizing contemporary research across organizational behavior, educational leadership, and digital transformation, the text prepares students to analyze the drivers of institutional excellence. Key thematic areas include the psychological dimensions of work—such as psychological ownership and positive deviant behavior—the strategic role of leadership in educational reform and knowledge management, and the critical transition toward digital ecosystems and e-learning. Furthermore, the book explores the intersection of sustainability and financial viability through models like the 'productive university' and sustainable consumption patterns. Through a diverse array of empirical case studies spanning the education, government, and hospitality sectors, students will develop the analytical tools necessary to evaluate how human capital, technological integration, and strategic governance converge to enhance organizational performance in a globalized context.

b) Prompt articles:

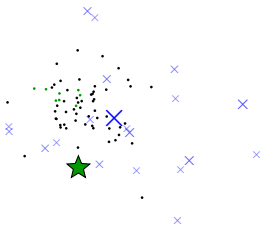
1) W4312239640 **الصورة الذهنية لجامعة الأزهر الإيجابية: رؤية ملموسة لبيئات تعليمية...**
2) W4391725010 **" أثر سلوكيات الانحراف الإيجابي لكفاءة أداء مهام العمل للعاملين...**
3) W2980345082 **تحليل العلاقة بين العمل الجماعى والمسئولية الاجتماعية (دراسة...**
4) W6893986298 **القائد التربوي ومهارات التسيير الفعال لتحسين الأداء التربوي...**
5) W4281946237 **إتجاهات الزوجة نحو أنماط الاستهلاك المستدام وعلاقتها بإتخاذ...**
6) W3161178110 **التحول الرقمى والتعلم عن بعد بالمملكة العربية السعودية خلال جائحة...**
7) W3135055911 **فاعلية برنامج تدريبي قائم على منصات التعلم الإلكتروني التفاعلية في...**
8) W3020471713 **الحاجات المهنية لدى المشرفين التربويين فى الأردن إلى الكفايات...**
9) W4293215330 **تبني جامعة البيضاء أنموذج الجامعة المنتجة لتنويع وتنمية مصادر...**
10) W4281251629 **تأثير الملكية النفسية الداعمة على سلوك العمل الإبداعى فى فنادق...**

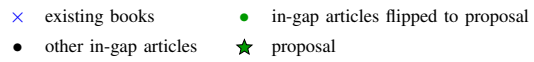
c) Prompt books:

1) W6968038398 **دانلود جزوه تجارت الكترونيکی**

I. `T14381` *Psychiatry, Mental Health, Neuroscience*

Topic has 140 books and 2321 articles; proposal covers 12.7% of the target gap; target gap size percentile = 100.0%.





a) Book proposal: **Television and Film Reception in the GDR: Methods and Perspectives**

This textbook provides a comprehensive introduction to the study of audiovisual media in the German Democratic Republic (GDR), designed to prepare students for research in media history and reception studies. By synthesizing empirical viewer data—specifically the 'Sofortresonanzen' (immediate response) reports—with historical and ideological analysis, the text equips students with the tools to examine the relationship between state-controlled broadcasting and audience behavior. Key modules cover the methodology of audience research in socialist contexts, the role of educational and entertainment films in shaping social norms, and the tension between official ideological goals and actual viewer reception. Through case studies of popular genres and specific program evaluations, students will learn to navigate the complex interplay of media policy, demographic variables, and cultural consumption within the specific socio-political framework of the GDR.

b) Prompt articles:

1) `W2917667297` **Sofortresonanzen DDR Fernsehen 19. Woche 1988 (SFR 19/88)**
2) `W6889779723` **Bereit für eine neue Zeit? Unterrichtsfilme der SBZ/DDR bis 1960**
3) `W2262119517` **Sofortresonanzen DDR Fernsehen 1. Woche 1984 (SFR 1/84)**
4) `W2917172812` **Sofortresonanzen DDR Fernsehen 34. Woche 1989 (SFR 34/89)**
5) `W2274865081` **Sofortresonanzen DDR Fernsehen 33. Woche 1984 (SFR 33/84)**
6) `W2917772989` **Sofortresonanzen DDR Fernsehen 28. Woche 1985 (SFR 28/85)**
7) `W2917104896` **Sofortresonanzen DDR Fernsehen 32. Woche 1988 (SFR 32/88)**
8) `W7025183538` **Unterhaltung, aber sicher! populäre Repräsentationen von Recht und Ordnung...**
9) `W2228917988` **Sofortresonanzen DDR Fernsehen 6. Woche 1982 (SFR 6/82)**
10) `W2917429225` **Sofortresonanzen DDR Fernsehen 3. Woche 1983 (SFR 3/83)**

c) Prompt books:

1) `W6945955271` **Film – Kino – Zuschauer: Filmrezeption. Film – Cinema – Spectator: Film...**
2) `W2480020764` **Rundfunkpolitik zwischen Sollen, Wollen und Können**
3) `W6927153470` **Ähnlichkeit**
4) `W6964124677` **Aktive Videoarbeit mit Jugendlichen mit geistiger Behinderung**
5) `W4251681977` **Medienwissenschaft**

J. T10215 Semantic Web and Ontologies

Topic has 620 books and 52487 articles; proposal covers 6.1% of the target gap; target gap size percentile = 100.0%.



a) Book proposal: **Foundations of the Semantic Web and Ontological Engineering: From Theory to Application**

This textbook provides a comprehensive introduction to the Semantic Web and the field of ontological engineering, designed to equip students with the theoretical and practical foundations necessary to engage with current research. Beginning with the fundamental shift from term-based to concept-based information representation, the text explores the architecture of the Semantic Web, the role of ontologies as shared conceptual schemes, and the technical standards that enable machine-readability, such as RDF and OWL. Students will be guided through the entire ontological engineering lifecycle—from knowledge elicitation and formalization to validation and implementation. Furthermore, the book examines critical themes of interoperability, Linked Open Data, and the integration of semantic technologies with Web 2.0. Through a variety of case studies in domains such as biomedicine, education, and tourism, the text demonstrates how to apply these frameworks to solve real-world problems in semantic search, automated web services, and linguistic annotation, bridging the gap between academic theory and industrial application.

b) Prompt articles:

1) W2338967774 **Hacia la primacía de los conceptos sobre los términos en los vocabularios...**
2) W1572648357 **Web Semântica: Conceitos Básicos e Tecnologias Associadas**
3) W2889614048 **Framework Basado en ODA para la Descripción y Composición de Servicios Web...**
4) W3185891010 **Hacia el uso efectivo de ontologías en aplicaciones informáticas**
5) W2552704197 **Buenas prácticas para la publicación de datos en la Web: La superación del...**
6) W1514905538 **Proposta de um mecanismo de busca baseado na web semântica para objetos de...**
7) W2239743315 **Integración de la web 2.0 y la web semántica mediante Mashups semánticos**
8) W2119562455 **¿De verdad sabes lo que quieres buscar? Expansión guiada visualmente de la...**
9) W6901482504 **Materialização da Web Semântica: um modelo de construção dinâmica de...**
10) W2994509800 **Primeras aproximaciones a la anotación lingüístico-ontológica de...**

c) Prompt books:

1) W812852241 **Annotation documentaire et peuplement d'ontologie à partir d'extractions...**
2) W583493747 **Ontological Landscapes**
3) W4285025634 **Proceedings Of The 1St Cuban Workshop On Semantic Web**
4) W2465821311 **Ontological Engineering: with examples from the areas of Knowledge...**
5) W2415161933 **From Semi-Formal Models to Formal Models**